

Remote Work Survey Analysis (2020)

Project Insights Documentation

Project Objective

Analyze how COVID-19 transformed remote work by studying employee demographics, productivity, work habits, barriers, experiences, and future preferences.

The analysis aims to answer:

- How productive are employees while working remotely?
- What factors influence productivity?
- What challenges do employees face?
- Is remote work sustainable after COVID-19?
- Which groups benefit the most from remote work?

Business Questions to Answer

1. Employee Demographics Analysis

Purpose:

Understand who participated in the survey.

Insights

- Age distribution
- Gender ratio
- Employment type
- Industry distribution
- Years of work experience
- Organization size
- State/Region distribution (if available)

Questions

- Which age group prefers remote work?
- Which industries adopted remote work the most?
- Do experienced employees adapt better?

2. Remote Work Adoption Analysis

Purpose:

Measure how widely remote work was adopted.

Insights

- Percentage working remotely
- Percentage working hybrid
- Percentage working on-site
- Days worked remotely
- Before vs During COVID comparison (if available)

Questions

- How much did remote work increase?
- Which industries shifted the fastest?

3. Productivity Analysis

One of the most important sections.

KPIs

- Average productivity score
- Productivity increase %
- Productivity decrease %
- Productivity unchanged %

Insights

- Does remote work improve productivity?
- Which employees are most productive?
- Does experience affect productivity?
- Which work setup gives maximum productivity?

Possible Visuals

- Productivity Distribution
- Productivity by Gender
- Productivity by Age
- Productivity by Industry
- Productivity Heatmap

4. Work-Life Balance Analysis

Questions

- Has work-life balance improved?
- Which employees experienced better balance?
- Does working remotely reduce stress?

KPIs

Average Work-Life Balance Score

Insights

- Balance by Gender
- Balance by Age
- Balance by Experience
- Balance by Industry

5. Working Hours Analysis

Questions

- Do employees work longer from home?
- Which group works the most hours?
- Does productivity increase with more hours?

KPIs

Average Daily Working Hours

Possible Relationships

Working Hours

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Productivity

↓

Satisfaction

6. Productivity Drivers

One of the strongest analytical sections.

Identify factors increasing productivity.

Possible Factors

- Quiet workspace
- Flexible timing
- Better internet
- No commuting
- Comfortable home office
- Family support

Questions

- Which factor contributes most?
- Which combination gives highest productivity?

7. Remote Work Barriers Analysis

This dimension table was created specifically for this.

Possible Barriers

- Internet issues
- Childcare
- Communication problems
- Lack of equipment
- Mental health
- Isolation

- Collaboration issues
- Home distractions

Questions

- Most common barrier?
- Which industries face the biggest challenges?
- Which age group struggles most?

KPIs

Top 10 Remote Work Barriers

8. Communication & Collaboration Analysis

Questions

- Is communication harder remotely?
- Which employees report collaboration issues?
- Does communication affect productivity?

Possible Relationship

Communication Score

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Team Collaboration

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Productivity

9. Remote Work Experience Analysis

Questions

- Overall experience score
- Employee satisfaction
- Comfort level
- Technology readiness

Insights

- Average Experience Rating
- Best Experience Group
- Worst Experience Group

10. Future of Remote Work

Very important business question.

Questions

- Do employees want remote work permanently?
- Hybrid or Full Remote?
- Return to Office?

KPIs

Preference Distribution

Possible Chart

Pie Chart

Remote

Hybrid

Office

11. Employee Satisfaction Analysis

Questions

- Overall satisfaction
- Satisfaction by work type
- Satisfaction by industry
- Satisfaction by gender

Relationship

Higher Satisfaction



Higher Productivity

12. Technology Readiness

Questions

- Do employees have enough equipment?
- Internet quality
- Software availability

Insights

Technology vs Productivity

Technology vs Satisfaction

13. Mental Health Analysis

Questions

- Does remote work reduce stress?
- Isolation effect
- Burnout indicators

Possible Relationship

Stress



Productivity



Work-Life Balance

14. Commute Time Saved

Business Value

Calculate average commute time saved.

Insights

Time Saved

↓

More Productivity

↓

Better Work-Life Balance

15. Remote Work Preference Analysis

Your table:

Dim_Productivity_Preference

Questions

- Why do employees prefer remote work?
- Why do employees dislike remote work?

Possible Reasons

- Flexibility
- Family
- Commute
- Productivity
- Health
- Office collaboration

16. Cross Analysis

This is where SQL shines.

Examples

Productivity vs Age

Does age influence productivity?

Productivity vs Gender

Who reports higher productivity?

Productivity vs Industry

Which industry benefits the most?

Productivity vs Work Experience

Does experience improve adaptation?

Productivity vs Work-Life Balance

Are happier employees more productive?

Productivity vs Remote Work Days

How many remote days maximize productivity?

Satisfaction vs Barriers

Do more barriers reduce satisfaction?

Technology vs Productivity

Does better equipment improve performance?

Work Hours vs Productivity

Do longer hours actually help?

17. Correlation Analysis (EDA)

Using Python

Correlation Matrix

Variables

- Productivity
- Satisfaction
- Working Hours
- Work-Life Balance
- Stress
- Experience

Output

Heatmap

Strong Positive Relationships

Strong Negative Relationships

19. Segmentation Analysis

Segment employees into groups

Examples

High Productivity + High Satisfaction

High Productivity + Low Satisfaction

Low Productivity + High Stress

Low Productivity + High Barriers

Excellent for Power BI filtering.

20. Executive Summary (Final Dashboard)

KPI Cards

- Total Employees Surveyed
- Average Productivity Score
- Average Satisfaction Score
- Average Work-Life Balance
- Average Working Hours

- Employees Preferring Remote Work
- Employees Preferring Hybrid Work
- Employees Preferring Office Work
- Top Productivity Driver
- Biggest Remote Work Barrier

Dashboard Pages

1. Executive Overview

- Overall KPIs
- Demographics
- Work Mode Distribution
- Key Insights

2. Employee Demographics

- Age
- Gender
- Industry
- Experience
- Organization Size

3. Productivity Analysis

- Productivity Scores
- Productivity Drivers
- Work Hours vs Productivity
- Productivity by Demographics

4. Remote Work Experience

- Satisfaction
- Work-Life Balance
- Experience Ratings
- Technology Readiness

5. Barriers & Challenges

- Top Barriers
- Communication Issues
- Mental Health
- Collaboration Challenges

6. Future of Remote Work

- Remote vs Hybrid vs Office Preferences
- Reasons Behind Preferences
- Business Recommendations